

PREDICTING TRAFFIC ACCIDENT SEVERITY IN BRAZIL

A Multiclass Machine Learning Approach with Imbalance Handling

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The Problem

33,000

deaths per year on
Brazilian roads

4%

of Brazil's GDP
(~R\$220 billion/yr)

10th

leading cause of
death nationally

Research has focused mainly on well-developed countries. Low- and middle-income countries, despite bearing a disproportionate share of the burden, remain understudied.

Research Questions

RQ1

When it comes to predicting the multiclass severity of accidents using macro-F1 and per-class metrics, can tree-based classifiers (Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost) outperform a Logistic Regression baseline?

RQ2

How does handling class imbalance affect the performance of these models, and how do the main methods of class imbalance handling compare?

RQ3

Which features are most important in predicting traffic accident severity?

Same Dataset, Same Three Blind Spots

Prior PRF studies — Scholz et al. (2024) and Gabardo et al. (2026) — share three limitations this work addresses.

Prior Work

Binary severity classification
(lethal / non-lethal)

Imbalance ignored, or only
one technique (SMOTE)

No explainability analysis
at the national level

This Study

**Three-class: without / injured /
dead victims**

**Seven imbalance methods
compared
systematically across 3 classifiers**

**Country-level TreeSHAP —
global + per-class + dependence
plots**

The Dataset

463,152

records

27

raw columns

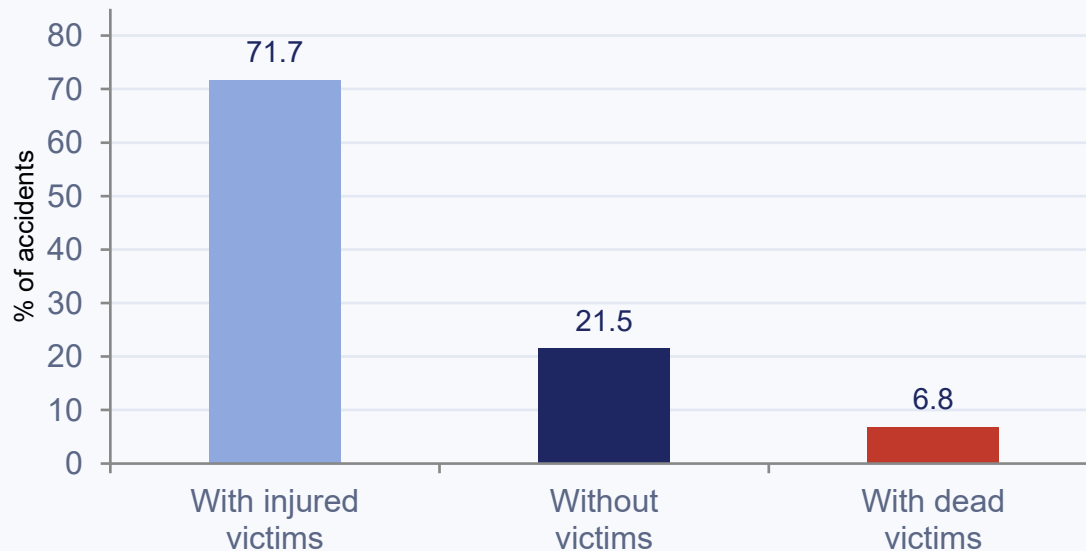
2017–2023

7 years, all of Brazil

PRF

Brazilian Federal
Highway Police (Kaggle)

Target: Victims_Condition — heavily imbalanced



A naive model predicting the majority class every time would look 71.7% accurate, while being useless for the 6.8% of cases that matter most.

Six-Phase Methodology

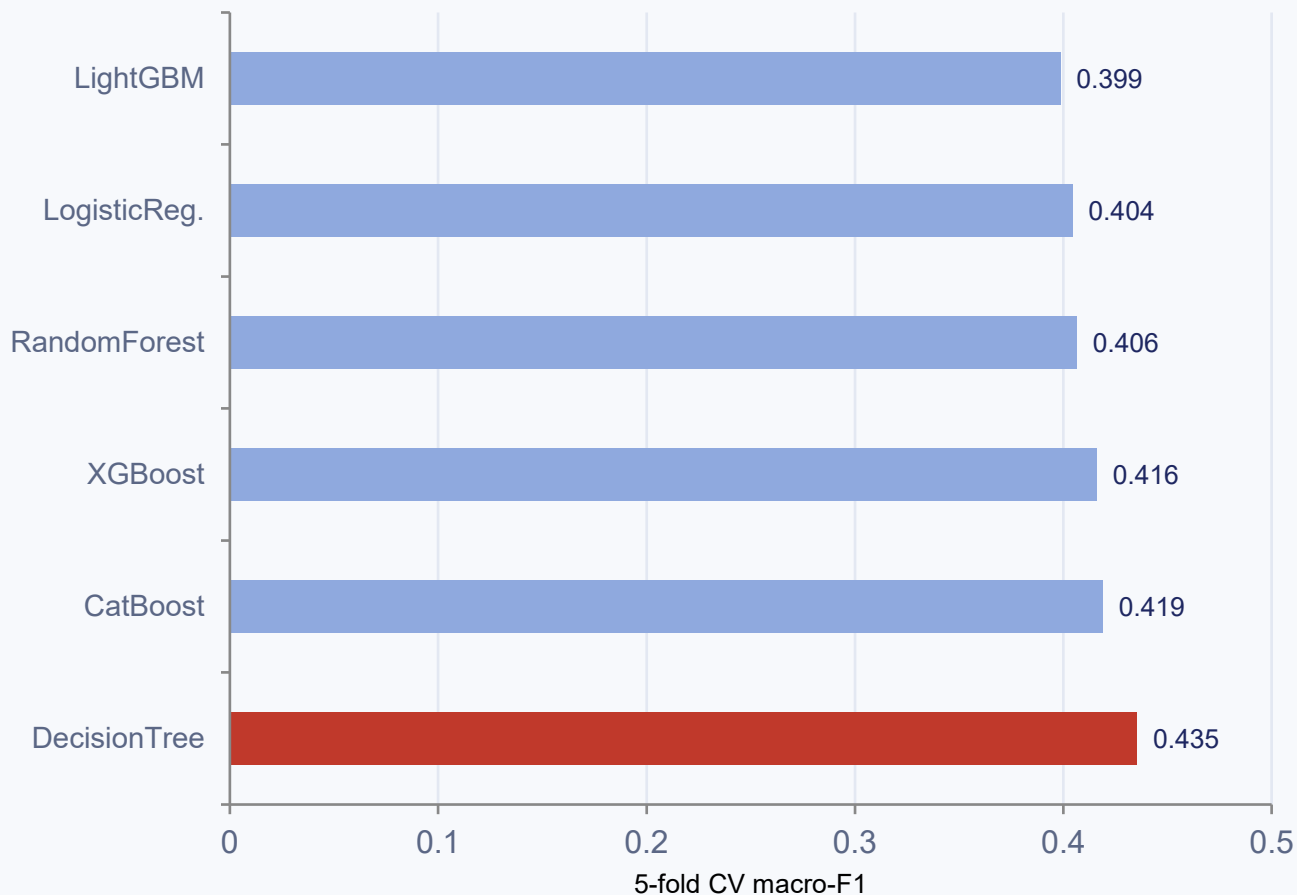
Phase 1 sets the bar: a DummyClassifier baseline of macro-F1 = 0.28 that every model must clear. Each phase narrows the field and feeds the next.



Primary metric: Macro-F1 — treats every class equally regardless of size

Secondary metric: MCC (Matthews Correlation Coefficient) — recommended for imbalanced multiclass data

Phase 2 — The Baseline Surprise



DecisionTree ranked first (0.4353) — ahead of every gradient booster.

That contradicts the literature (Grinsztajn 2022; Shwartz-Ziv & Armon 2022), which predicts boosters should dominate tabular data.

Two sanity checks performed:

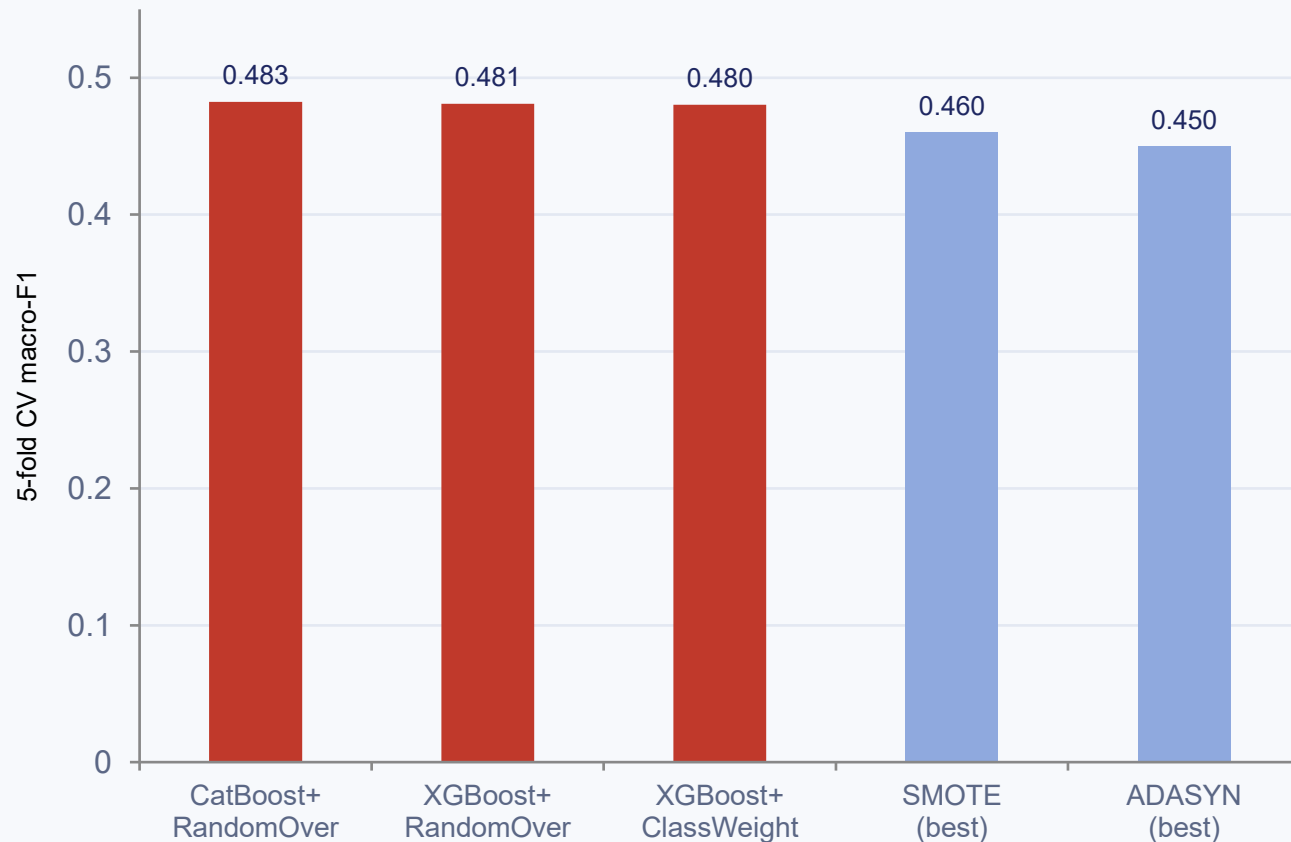
Full-train → sealed test set: 0.4397 (no drop)

Depth capped at 20, re-CV:

0.4280 (nearly identical)

Conclusion: not overfitting — the categorical features carry a strong enough signal for a single tree to capture it.

Phase 3 — Simple Beats Sophisticated



21 combinations tested (3 classifiers × 7 imbalance methods). The methods most recommended in the literature (SMOTE, ADASYN) underperformed simple oversampling and class weighting by around 0.02 macro-F1.

Why?

After one-hot encoding, SMOTE interpolates between rows — creating impossible records (e.g. 60% Northeast + 40% Southeast, which can't happen in reality).

RandomOversampler just duplicates real rows, so it never creates nonsense.

Phase 4 — Tuning Confirms the Winner

XGBoost + RandomOver

0.5188

CV macro-F1 — best of 50 Optuna trials (trial 43)

Optuna ran 150 trials total (3 combinations × 50) — about 10 hours of Bayesian optimisation, overtaking CatBoost + RandomOver, which had led after Phase 3.

Gain over Phase 3

0.4811 → 0.5188

+0.038 macro-F1 from tuning alone

Limitation flagged early

max_depth saturated at its upper bound (10) across all three Optuna studies — the search range was possibly too narrow. Carried forward to Future Work.

Final Test-Set Results

Macro-F1

0.28 → 0.52
baseline final model

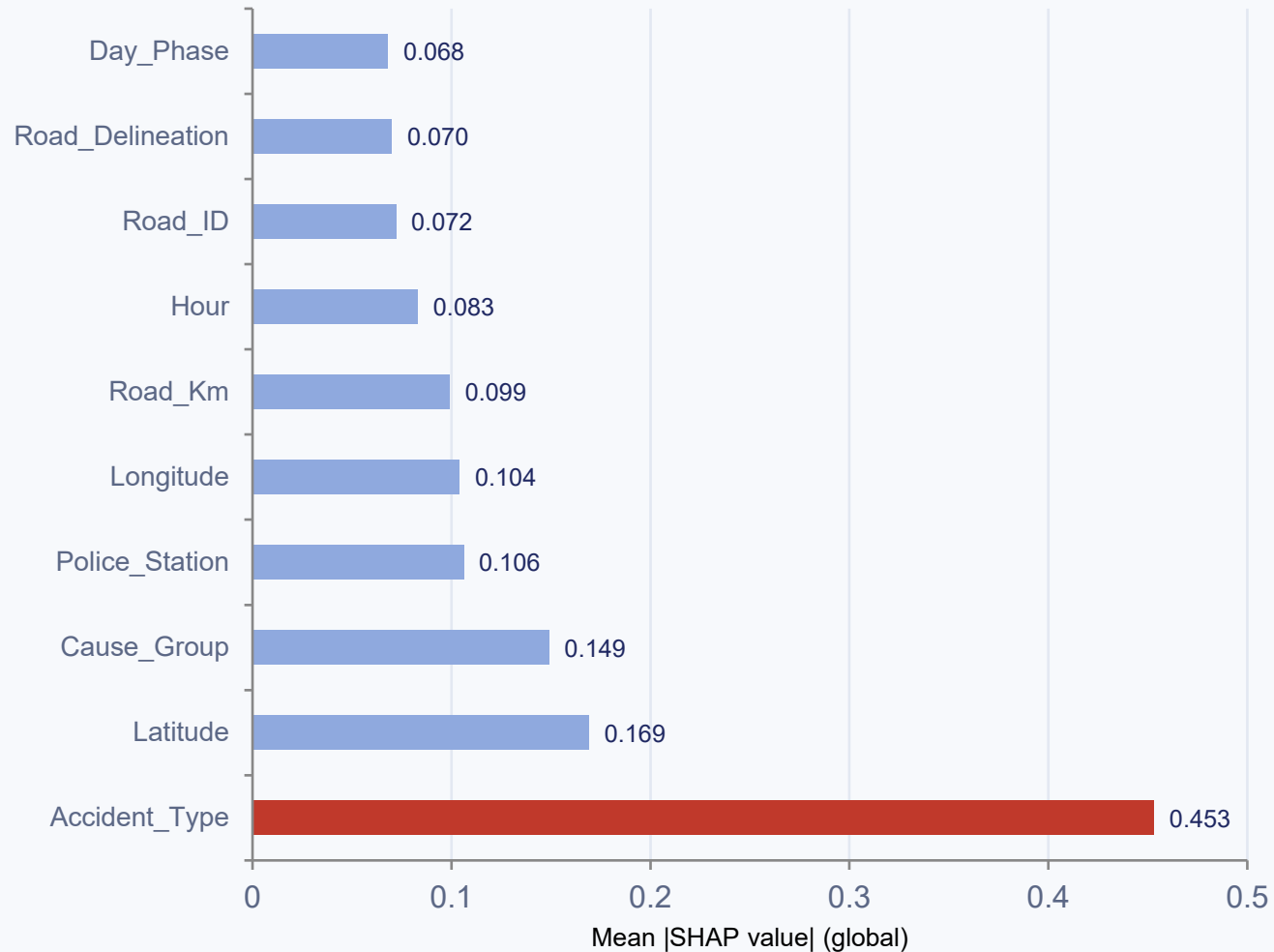
Fatal-Class Recall

22.4% → 50.1%
baseline final model

More than doubled: 1,403 of 6,269 fatal cases caught by the Phase 2 baseline vs. 3,141 of 6,269 by the final tuned model.

The honest trade-off: some injured cases get misclassified as fatal. In a safety context, a false alarm on a fatal case is a far smaller cost than missing one — and CV (0.5188) vs. test (0.5194) macro-F1 stayed within 0.0006, showing no overfitting.

What Actually Drives Severity



Accident_Type dominates —

Around 3× the second-ranked feature (Latitude). Police_Station ranks 4th as a geographic proxy, not a causal factor.

Day_Phase

Top 5 for fatal accidents — but drops to 14th for injured. Time of day matters specifically for whether a crash becomes fatal.

Weather_Condition

Only reaches top 5 for the "without victims" class — suggesting drivers take more care in bad weather, rather than weather directly causing severe crashes.

Contributions, Limitations & Future Work

Contributions

- First multi-class severity study on the full national PRF dataset (not binary, not regional)
- Systematic comparison of 7 imbalance-handling methods across 3 classifiers (21 combinations)
- First country-level TreeSHAP explainability analysis on PRF data
- Finding: simple resampling (RandomOver, ClassWeight) outperforms SMOTE/ADASYN on high-cardinality OHE data

Limitations

- LightGBM never tuned in Phase 4, its true potential is untested
- max_depth saturated at its upper bound (10) in every winning Optuna trial
- Police_Station ranks 4th in SHAP but is a geographic proxy, not a causal factor
- Standard SMOTE isn't built for one-hot categorical features (SMOTE-NC would be more appropriate)

Future Work

- Widen the max_depth/depth search range (10 → 15–20) and test SMOTE-NC
- Tune LightGBM to see if it changes the Phase 2 ranking
- Add spatio-temporal risk scoring (hour + road segment + historical pattern)
- Validate on 2024–2025 PRF data to test deployment readiness

Thank You

Questions & Discussion

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GitHub: <https://github.com/Aang1214/Dissertation>

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